

Single-Cell RNA Sequencing Analysis of Leukemia versus Normal Hematopoietic Cells

A Comparative Evaluation of Three CellTypist Annotation Models for Differential Gene Expression



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Single-Cell RNA-Seq Analysis Report

Abstract

This report presents a single-cell RNA sequencing (scRNA-seq) analysis comparing a Leukemia sample (971 cells) and a Normal sample (760 cells), sequenced on the Illumina HiSeq 4000 platform. After quality control, normalization, highly variable gene selection, PCA, and Leiden clustering, cells were annotated into biological cell types using three independent CellTypist reference models. Differential gene expression (DGE) between Leukemia and Normal cells was then evaluated within each annotated cell type (Wilcoxon rank-sum test; $|\log_2FC| \geq 1$, adjusted $p < 0.05$). The hematopoietic stem/progenitor (HSC-MPP) population was the only cluster testable under all three models.

Objective

- Process and quality-control raw single-cell expression data from Leukemia and Normal samples.
- Cluster cells and annotate cell types using three independent CellTypist reference models.
- Identify differentially expressed genes between Leukemia and Normal cells within each annotated cell type.
- Compare the three annotation models and determine which best describes the differential gene expression between conditions.

Introduction

Single-cell RNA sequencing enables gene expression profiling at individual-cell resolution, resolving cellular heterogeneity that bulk RNA-seq averages out. This is especially useful in leukemia, where malignant transformation arises within specific hematopoietic stem and progenitor populations while other blood lineages remain comparatively normal. Comparing a Leukemia sample against a healthy Normal reference at single-cell resolution localizes disease-associated transcriptional changes to the specific populations in which they occur.

A key step in this workflow is automated cell type annotation, performed here using CellTypist, which assigns a label to each cell by comparison against a pre-trained reference model. Because reference models differ in granularity and biological scope, the same dataset can be annotated differently depending on the model applied. This report evaluates three CellTypist models side by side and compares the resulting differential expression outcomes, to identify which model most reliably captures the transcriptional differences between Leukemia and Normal cells.

Methodology

Dataset and Quality Control

Two 10x Genomics-format samples were used: Normal (760 cells) and Leukemia (971 cells), each with 36,601 genes. The samples were concatenated (1,731 cells total) with sample identity retained for group comparisons. Per-cell QC metrics (total counts, genes detected, counts-per-gene, %mitochondrial, %ribosomal) were computed, and cells were kept only within the 0.001-0.999 quantile range for counts/genes/complexity, with mitochondrial content $< 15\%$ and ribosomal content $< 50\%$. This reduced the dataset to 1368 cells (363 cells, 20.97%, removed).

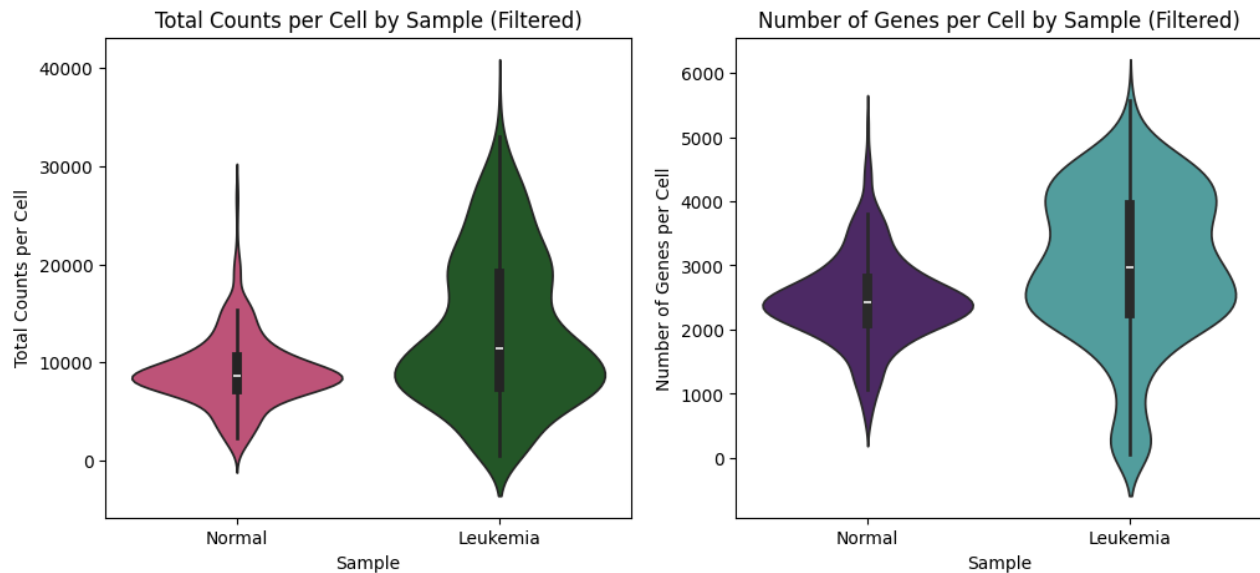


Figure 1. Total counts and gene counts per cell, by sample, after QC filtering.

Normalization, Clustering and Annotation

Filtered counts were library-size normalized (target sum = 1e4) and log1p-transformed. The top 2,000 highly variable genes were used for PCA (~50 PCs captured approximately 80% of the variance) and neighborhood-graph construction. Leiden clustering was performed at a resolution of 0.7, resulting in eight transcriptionally distinct clusters (clusters 0–7), which were subsequently annotated using CellTypist majority voting. Cell type identities were assigned using three independent CellTypist reference models. Model 1 (Adult_cHSPCs_Illumina.pkl) predicted seven hematopoietic progenitor populations, Model 2 (Immune-all-low.pkl) predicted eight broader hematopoietic populations, and Model 3 (Immune-all-high.pkl) predicted six major hematopoietic populations. Differential expression analysis was then performed separately within each annotated cell type by comparing Leukemia and Normal cells whenever both groups contained at least three cells.

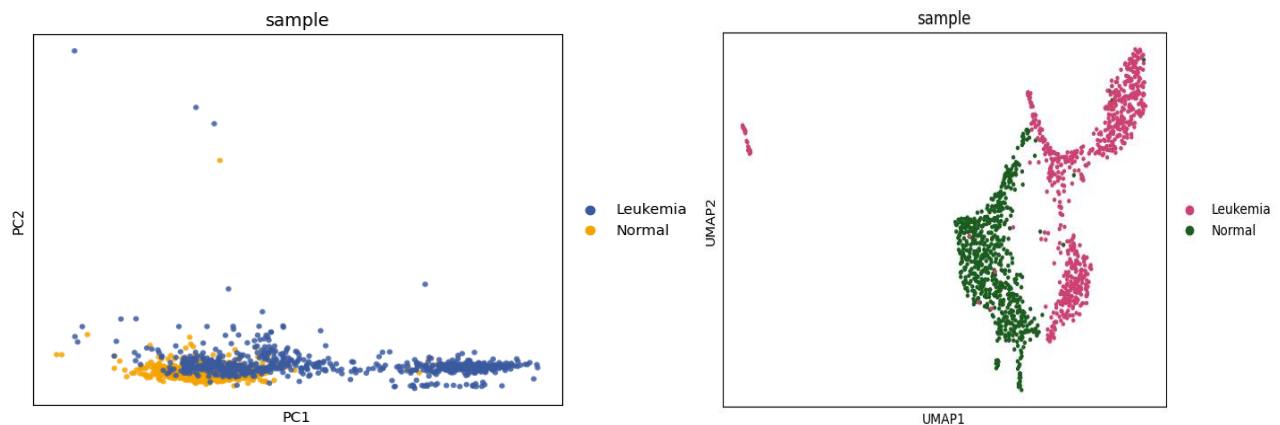


Figure 2. Left: PCA embedding colored by sample. Right: UMAP colored by Leiden cluster (resolution 0.7).

Differential Expression Analysis

For each annotated cell type, under each of the three models, Leukemia cells were compared against Normal cells (reference) using `sc.tl.rank_genes_groups` (Wilcoxon test, log-normalized data). Genes were called significant at adjusted $p < 0.05$ and $|\log_2FC| \geq 1$. Cell types with only one condition present, or fewer than three cells in either group, were excluded from testing.

Results

Cell Type Annotation (Model 3)

After quality control, the three CellTypist models produced different cell-type distributions.

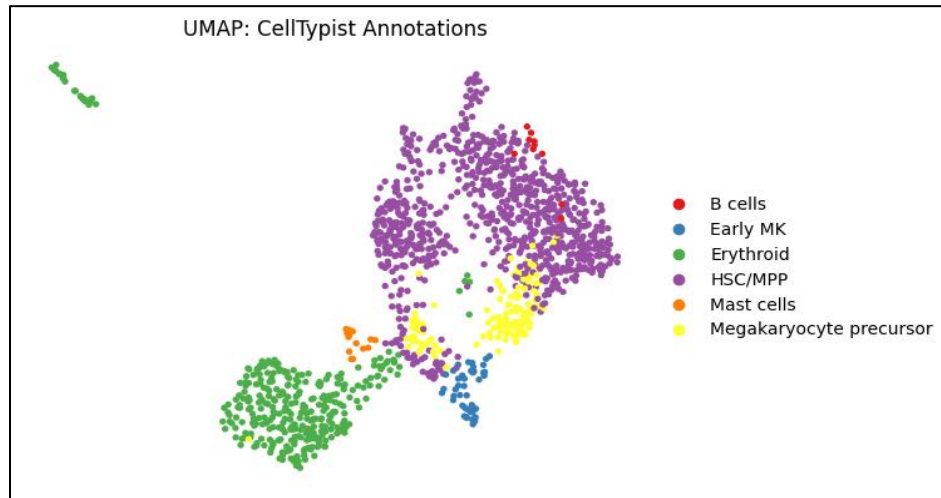


Figure 3

Model 3 produced a biologically coherent CellTypist annotation of the dataset. The UMAP was dominated by an HSC/MPP population, while erythroid cells formed a clearly separated cluster and megakaryocyte precursor and Early MK populations occupied neighboring regions consistent with hematopoietic differentiation. Smaller Mast-cell and B-cell populations were also identified as distinct clusters. Overall, the spatial separation and relative organization of the major populations indicate that Model 3 provides a clear and biologically interpretable representation of the underlying hematopoietic cell populations, supporting its selection as the preferred annotation model among the three models tested.

Cell Type(Model 2)	Cell Count
Erythroid	340
HSC/MPP	836
Early MK	126
Megakaryocyte precursor	45
Mast cells	15
B cells	10

Table 1. CellTypist Model 3 annotation counts across the 323 filtered cells.

Differential Expression Outcomes Across Models

Differential expression analysis was performed separately within each annotated cell type using the Wilcoxon rank-sum test. Cell populations containing fewer than three cells in either Normal or Leukemia groups were excluded. Table 2 summarizes every cell type for which testing was actually performed.

Model	Cell Type	Normal (n)	Leukemia (n)	Significant DEGs
Model 1	HSC/MPP	566	228	244
Model 1	Megakaryocyte precursor	15	64	9
Model 1	GMPL	11	22	0
Model 2	HSC-MPP	587	223	239
Model 2	Mid Erythroid	5	183	0
Model 3	HSC_MPP	556	280	289
Model 3	MKP	90	36	82
Model 3	Erythroid	7	329	0

Table 2. DGE testing outcome per cell type/model (Wilcoxon, $\text{padj} < 0.05$, $|\log_2\text{FC}| \geq 1$)

Direct Comparison on the HSC-MPP Population

HSC-MPP was the only cell type annotated with sufficient cells for testing under all three models, making it the most reliable basis for comparison (Figure 4).

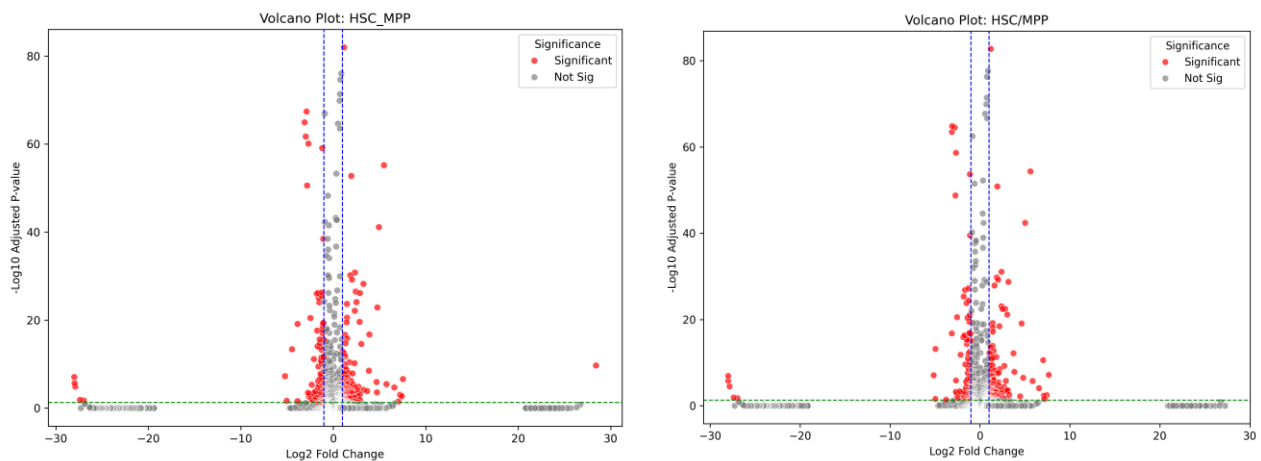
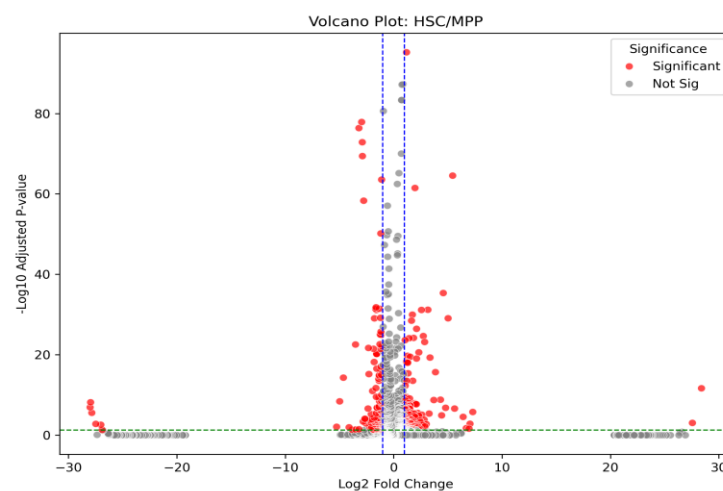


Figure 4. HSC-MPP volcano plots.: Model 1 (244 sig. genes).

Model 2 (239 sig. genes).



Model 3 (289 sig. genes).

Top genes of HSC-MPP are listed in Table 3.

Gene	log2 Fold Change	Adjusted p-value	Direction
RPS26	1.18	6.2×10^{-7}	Up in Leukemia
HBB	5.44	2.8×10^{-7}	Up in Leukemia
RPS23	-0.03	1.3×10^{-5}	Down in Leukemia
RPL7	-0.05	0.1×10^{-5}	Down in Leukemia

Table 3. Top significant DEGs identified by Model 3 in HSC-MPP (full 289-gene list in the accompanying CSV).

Discussion: Which Model Best Describes the DEGs

Among the three CellTypist annotation models, **Model 3 (Immune-all-high)** provided the strongest differential expression results for the HSC/MPP population, identifying **289 significant genes**, compared with **244** in Model 1 and **239** in Model 2.

Model 3 also produced a simpler annotation by grouping erythroid cells into a single population while retaining a large HSC/MPP compartment (836 cells), providing greater statistical power for differential expression testing. In contrast, Model 1 divided progenitor populations into several smaller clusters (MKP, MEBEMP-L, GMP-L, ERY-P), which allowed additional cell-type-specific DE analysis but reduced statistical power in some rare populations. Model 2 produced an intermediate annotation strategy by separating early, mid, and late erythroid populations.

For rare cell populations, all three models showed limited differential expression because Normal cells were scarce after filtering, resulting in many comparisons being skipped or producing no significant genes. Consequently, the most reliable biological comparison across all models remained the HSC/MPP population.

Output of model3:

```
Processing cell type: HSC/MPP
sample
Normal      556
Leukemia    280
Name: count, dtype: int64
DEGs saved for HSC/MPP: total=36601, significant=289
Volcano plot saved for HSC-MPP
```

```
Processing cell type: Megakaryocyte precursor
sample
Normal      90
Leukemia    36
Name: count, dtype: int64
DEGs saved for Megakaryocyte precursor: total=36601, significant=82
Volcano plot saved for Megakaryocyte_precursor
```

```
Processing cell type: B cells
sample
Normal     10
Name: count, dtype: int64
Skipping B cells: only one sample group present, cannot run DE
```

```
Processing cell type: Early MK
sample
```

```
Leukemia      33
Normal        12
Name: count, dtype: int64
DEGs saved for Early MK: total=36601, significant=3
Volcano plot saved for Early_MK
```

```
Processing cell type: Erythroid
sample
Leukemia      329
Normal         7
Name: count, dtype: int64
DEGs saved for Erythroid: total=36601, significant=0
Volcano plot saved for Erythroid
```

```
Processing cell type: Mast cells
sample
Leukemia      14
Normal         1
Name: count, dtype: int64
Skipping Mast cells: one or more groups has too few cells (<3)
```

Conclusion

This analysis processed and quality-controlled a Leukemia versus Normal single-cell dataset, clustered the cells, and annotated them using three independent CellTypist models. Differential expression testing was statistically meaningful only for the HSC-MPP population, adequately represented in Leukemia and Normal groups under all three models. Model 3 produced the most statistically robust and biologically interpretable DEG set (289 significant genes, including immune and cell-cycle markers) and is recommended as the most suitable annotation model for describing differential expression in this dataset.